Status	Finished
Started	Tuesday, 20 October 2026, 9:51 PM
Completed	Tuesday, 20 October 2026, 9:55 PM
Duration	3 mins 59 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are split again into smaller sublists
b. They are discarded
c. They are compared element by element
d. They are combined to form a single sorted list

The correct answer is: They are combined to form a single sorted list

Question **2**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. sorted()
b. arrange()
c. sort()
d. order()

The correct answer is: sort()

Question **3**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Sorting
- b. Searching
- c. Merging
- d. Rearranging

The correct answer is: Sorting

Question **4**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Distribution
- b. Extraction
- c. Selection
- d. Insertion

The correct answer is:
Insertion

Question **5**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Linear Search
- b. Binary Search
- c. Bubble Sort
- d. Merge Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort cannot handle duplicate elements
b. Bubble Sort does not guarantee sorted order
c. Bubble Sort is difficult to implement
d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **7**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It sorts data using a single pass
b. It does not require recursion
c. It is based on the divide-and-conquer approach
d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach

Question **8**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Quick Sort
b. Merge Sort
c. Linear Search
d. Bubble Sort

The correct answer is: Merge Sort

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Arranging data in a specific order
- b. Removing duplicates from a list
- c. Merging two datasets
- d. Finding a specific element in a list

The correct answer is: Arranging data in a specific order

Question **10**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Dynamic programming
- b. Greedy method
- c. Backtracking
- d. Divide and conquer

The correct answer is:
Divide and conquer

Question **11**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It complicates the selection of items
- b. It makes it easier to select items based on their relationship to the rest of the items
- c. It makes the data unsorted
- d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **12**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It only works on small datasets
b. It avoids the need for recursion
c. It allows for efficient parallel processing and sorting of data
d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **13**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Decreases the efficiency of algorithms
b. Makes data harder to manage
c. Provides a basis for other algorithms to work efficiently
d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Insertion Sort
b. Bubble Sort
c. Merge Sort
d. Quick Sort

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It increases the number of comparisons needed
b. It has no effect on the search operation
c. It allows for faster searching
d. It makes the search operation slower

The correct answer is: It allows for faster searching